

Theoretical Quantum Optics: Problem Sheet 7

(for 23.06.2025)

20. Collective light-matter interactions (2P)

An ensemble of N two-level atoms is coupled collectively to a fast decaying cavity mode and is driven by an external driving field. By treating the cavity as a Markovian reservoir, the dynamics of the atoms can be described by the master equation

$$\dot{\rho} = -i\Omega N[J_x, \rho] + \frac{\Gamma}{2} (2J_- \rho J_+ - J_+ J_- \rho - \rho J_+ J_-), \quad (1)$$

where $J_x = (J_- + J_+)/2$ and $J_{\pm} = \sum_{i=1}^N \sigma_{\pm}^i$ are the collective raising and lowering operators with

$$J_{\pm}|j, m\rangle = \sqrt{(j \mp m)(j \pm m + 1)}|j, m \pm 1\rangle. \quad (2)$$

We assume that the dynamics is constrained to the maximal angular momentum subspace with $j = N/2$.

- (a) Write a Python program to simulate the dynamics and the steady state of master equation (1).
- (b) Starting from the fully excited state $|ee \dots ee\rangle = |j = N/2, m = N/2\rangle$, simulate the decay dynamics for $\Omega = 0$. Plot the time evolution for the normalized population inversion $\langle J_z \rangle(t)/N$ and the emitted photon current $I_{\text{ph}}(t) = \Gamma \langle J_+ J_- \rangle(t)/N$ as a function of the normalized time $\tau = t\Gamma N$ for $N = 1, 10, 50$.
- (c) Evaluate the steady-state expectation values of $\langle J_z \rangle/N$ and $\langle J_y \rangle/N$ as a function of Ω/Γ and for $N = 1, 10, 50$. Show that as N increases, the steady state undergoes a sharper and sharper transition at a critical value of $\Omega/\Gamma = 0.5$.

Hint: Define the Liouville operator as a sparse matrix to reduce memory and computation time. Try out slightly higher values of N to see how big of a system you can still simulate in a reasonable time.

21. Quantum Langevin equation for a two-level atom (2P)

Consider a driven two-level atom with Hamiltonian

$$H = -\hbar\Delta|e\rangle\langle e| + \frac{\hbar\Omega}{2}\sigma_x. \quad (3)$$

The atom decays into a zero-temperature bath with a rate Γ and dephases due to a coupling to a second bath with rate Γ_{ϕ} . The corresponding master equation has been discussed in Sec. 4.2.3.

- (a) Derive the QLEs for the operators $\sigma_-(t)$, $\sigma_+(t)$ and $\sigma_z(t)$.

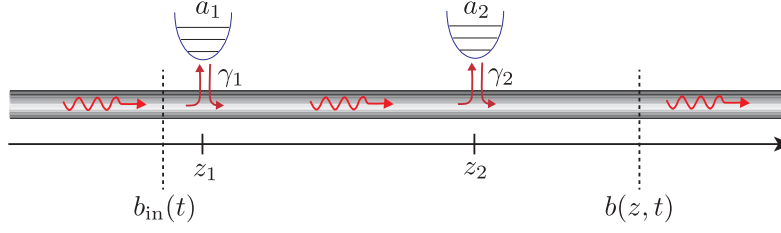


Abbildung 1: Sketch of a cascaded quantum system.

- (b) Show that the corresponding equations for the expectation values of these operators reproduce the correct optical Bloch equations.
- (c) Use the product rule to verify explicitly that

$$\partial_t(\sigma_+(t)\sigma_-(t)) = \partial_t\sigma_{ee}(t), \quad (4)$$

where $\sigma_{ee} = |e\rangle\langle e|$.

22. Cascaded quantum systems (2P)

Consider two harmonic oscillators, which are coupled to a unidirectional waveguide at positions z_1 and z_2 , as shown in Fig. 1. The Hamiltonian for the full system is given by

$$H = \sum_{i=1}^2 \hbar\omega_0 a_i^\dagger a_i + \int_{\omega_0-\Delta}^{\omega_0+\Delta} d\omega \hbar\omega b_\omega^\dagger b_\omega + H_{\text{int}}, \quad (5)$$

where $[b_\omega, b_{\omega'}^\dagger] = \delta(\omega - \omega')$ and the interaction Hamiltonian is

$$H_{\text{int}} = i \sum_{i=1}^2 \sqrt{\frac{\gamma_i}{2\pi}} \left[b^\dagger(z_i) a_i - b(z_i) a_i^\dagger \right], \quad b(z, t) = \frac{1}{\sqrt{2\pi}} \int d\omega e^{i\omega z/c} b_\omega(t). \quad (6)$$

- (a) Derive the QLEs for the operators a_1 and a_2 and express them in terms of the incoming field

$$b_{\text{in}}(t) = \lim_{\epsilon \rightarrow 0} b(z_1 - \epsilon, t). \quad (7)$$

- (b) Express the outgoing field $b(z, t)$ at position $z > z_2$ in terms of the $a_i(t)$ and $b_{\text{in}}(t)$.

21. Quantum Langevin equation for a two-level atom (2P)

Consider a driven two-level atom with Hamiltonian

$$H = -\hbar\Delta|e\rangle\langle e| + \frac{\hbar\Omega}{2}\sigma_x. \quad (3)$$

The atom decays into a zero-temperature bath with a rate Γ and dephases due to a coupling to a second bath with rate Γ_ϕ . The corresponding master equation has been discussed in Sec. 4.2.3.

(a) Derive the QLEs for the operators $\sigma_-(t)$, $\sigma_+(t)$ and $\sigma_z(t)$.

• Adding dephasing processes (=frequency fluctuations): $c \equiv \sigma_z$

$$\dot{\rho} = \dots + \frac{\Gamma_\phi}{4} (2\sigma_z \rho \sigma_z - \sigma_z^2 \rho - \rho \sigma_z^2)$$

↳ translates to decay in coherences but doesn't affect populations
1. the other

section 4.2.3

$$H_s = \hbar\omega_{eg}|e\rangle\langle e| + \frac{\hbar\Omega}{2}(\sigma_+ e^{-i\omega_L t} + \sigma_- e^{i\omega_L t}), \quad H_B = \sum_k \hbar\omega_k a_k^\dagger a_k$$

Remark: We can change into a rotating frame with respect to ω_L without affecting the dissipation terms:

$$H_s \rightarrow H_s = -\hbar\Delta|e\rangle\langle e| + \frac{\hbar\Omega}{2}\sigma_x$$

$$H_{\text{int}} = \hbar \sum_k (g_k \sigma_+ a_k + g_k^* \sigma_- a_k^\dagger) \equiv \hbar (cB^\dagger + c^\dagger B)$$

$$\Rightarrow c \equiv \sigma_-, \quad B \equiv \sum_k g_k a_k$$

With these identifications and assuming $T=0$, we obtain:

$$\dot{\rho}_s = -\frac{i}{\hbar}[H_s, \rho_s] + \frac{\Gamma}{2}(2\sigma_- \rho_s \sigma_+ - \sigma_+ \sigma_- \rho_s - \rho_s \sigma_+ \sigma_-)$$

where $\Gamma = 2\text{Re} \int_0^\infty d\tau \langle B(\tau)B^\dagger(0) \rangle e^{i\omega_{eg}\tau}$ is the atomic decay rate.

$$\begin{aligned} \text{Decay rate: } \Gamma &= 2\text{Re} \sum_k \int_0^\infty d\tau |g_k|^2 e^{i(\omega_{eg}-\omega_k)\tau} \\ &= 2\text{Re} \int_0^\infty d\omega |g(\omega)|^2 \rho(\omega) \int_0^\infty d\tau e^{i(\omega_{eg}-\omega)\tau} \\ &= \pi \delta(\omega - \omega_{eg}) \\ &= 2\pi |g(\omega_{eg})|^2 \rho(\omega_{eg}) \equiv \Gamma_{\text{Wigner-Weisskopf}} \end{aligned}$$

General open quantum system:

$$H = H_S + H_B + H_{\text{int}}$$

bandwidth large compared to system dynamics

• Bosonic reservoir: $H_B = \int_{\omega_0-\Delta}^{\omega_0+\Delta} d\omega \hbar\omega b_\omega^\dagger b_\omega, \quad [b_\omega, b_{\omega'}^\dagger] = \delta(\omega - \omega')$

$b_\omega \rightarrow b_\omega e^{i\omega\tau}$

• System-bath coupling: $H_{\text{int}} = i\hbar \int_{\omega_0-\Delta}^{\omega_0+\Delta} d\omega g(\omega) (c b_\omega^\dagger - c^\dagger b_\omega)$

↑
jump operator

5.1.1) Derivation of the QLE

For an arbitrary system operator $O_S(t)$ we obtain (Heisenberg EOM)

$$\dot{O}_S = \frac{i}{\hbar}[H, O_S] = \frac{i}{\hbar}[H_S, O_S] - \int d\omega g(\omega) (b_\omega^\dagger [c, O_S] - [c^\dagger, O_S] b_\omega)$$

$$\dot{\sigma}_-(t) = \frac{i}{\hbar} [H, \sigma_-(t)]$$

decay into bath: $H_{\text{int}} = \hbar \sum_k (g_k \sigma_+ a_k + g_k^* \sigma_- a_k^\dagger)$

$$= \hbar (cB^\dagger + c^\dagger B)$$

$$c = \sigma_-, \quad B = \sum_k g_k a_k$$

$$H = H_S + H_{\text{int } T=0 \text{ bath}} + H_{\text{dephasing}}$$

• Relaxation dynamics for $\Omega = 0$:

Remember: $\langle \dot{O} \rangle(t) = \frac{i}{\hbar} \langle [H, O] \rangle + \sum_k \frac{\Gamma_k}{2} \langle [c_k^\dagger, O] c_k + c_k^\dagger [O, c_k] \rangle$

-) $c_{1,2} = \{\sigma_-, \sigma_+\}, \quad \Gamma_{1,2} = \{\Gamma(N_{\text{th}} + 1), \Gamma N_{\text{th}}\}$

-) $O \equiv \sigma_z: \quad [\sigma_-, \sigma_z] = 2\sigma_-, \quad [\sigma_+, \sigma_z] = -2\sigma_+$

$$\Rightarrow \langle \dot{\sigma}_z \rangle = -\Gamma(2N_{\text{th}} + 1) \langle \sigma_z \rangle - \Gamma$$

$t \rightarrow \infty \Rightarrow \langle \dot{\sigma}_z \rangle = 0$ (steady state)

$$H_S = -\hbar\Delta|e\rangle\langle e| + \frac{\hbar\Omega}{2}\sigma_x$$

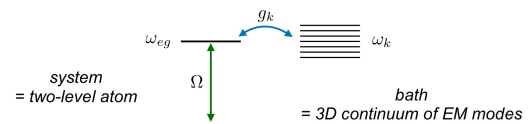
With these identifications and assuming $T=0$, we obtain:

$$\dot{\rho}_s = -\frac{i}{\hbar}[H_s, \rho_s] + \frac{\Gamma}{2}(2\sigma_- \rho_s \sigma_+ - \sigma_+ \sigma_- \rho_s - \rho_s \sigma_+ \sigma_-)$$

where $\Gamma = 2\text{Re} \int_0^\infty d\tau \langle B(\tau)B^\dagger(0) \rangle e^{i\omega_{eg}\tau}$ is the atomic decay rate.

$$\begin{aligned} \text{Decay rate: } \Gamma &= 2\text{Re} \sum_k \int_0^\infty d\tau |g_k|^2 e^{i(\omega_{eg}-\omega_k)\tau} \\ &= 2\text{Re} \int_0^\infty d\omega |g(\omega)|^2 \rho(\omega) \int_0^\infty d\tau e^{i(\omega_{eg}-\omega)\tau} \\ &= \pi \delta(\omega - \omega_{eg}) \\ &= 2\pi |g(\omega_{eg})|^2 \rho(\omega_{eg}) \equiv \Gamma_{\text{Wigner-Weisskopf}} \end{aligned}$$

4.2.3) Example: the driven TLA



• bare system & bath Hamiltonians (RWA):

$$H_s = \hbar\omega_{eg}|e\rangle\langle e| + \frac{\hbar\Omega}{2}(\sigma_+ e^{-i\omega_L t} + \sigma_- e^{i\omega_L t}), \quad H_B = \sum_k \hbar\omega_k a_k^\dagger a_k$$

• interaction Hamiltonian:

$$H_{\text{int}} = \hbar \sum_k (g_k \sigma_+ a_k + g_k^* \sigma_- a_k^\dagger) \equiv \hbar (cB^\dagger + c^\dagger B)$$

$$\Rightarrow c \equiv \sigma_-, \quad B \equiv \sum_k g_k a_k$$

• Adding dephasing processes (=frequency fluctuations): $c \equiv \sigma_z$

$$\dot{\rho} = \dots + \frac{\Gamma_\phi}{4} (2\sigma_z \rho \sigma_z - \sigma_z^2 \rho - \rho \sigma_z^2)$$

↳ translates to decay in coherences but doesn't affect populations

Quantum Langevin Equation (QLE):

$$\dot{O}_S = \frac{i}{\hbar} [H_S, O_S] - [O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{in}(t) \right) - \left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{in}^\dagger(t) \right) [c, O_S]$$

$\sigma_- (t)$:

$$H_S = -\hbar \Delta |e\rangle\langle e| + \frac{\hbar\Omega}{2} \sigma_x$$

$$\begin{aligned} [H_S, \sigma_-] &= -\hbar \Delta [|e\rangle\langle e| (t), \sigma_- (t)] + \frac{\hbar\Omega}{2} [\sigma_x (t), \sigma_- (t)] \\ &= -\hbar \Delta e^{iHt} [|e\rangle\langle e|, |g\rangle\langle g|] e^{-iHt} + \frac{\hbar\Omega}{2} e^{iHt} [(|g\rangle\langle e| + |e\rangle\langle g|), |g\rangle\langle e|] e^{-iHt} \\ &= \hbar \Omega \sigma_- (t) - \frac{\hbar\Omega}{2} \sigma_z (t) \end{aligned}$$

$$|e\rangle\langle e| |g\rangle\langle g| - |g\rangle\langle g| |e\rangle\langle e| = 0 - |g\rangle\langle g| = -|g\rangle\langle g| = -\sigma_-$$

$$[|g\rangle\langle g|, |g\rangle\langle g|] = 0$$

$$[|e\rangle\langle g|, |g\rangle\langle e|] = |e\rangle\langle g| |g\rangle\langle e| - |g\rangle\langle e| |e\rangle\langle g| = |e\rangle\langle e| - |g\rangle\langle g| = -\sigma_z$$

for the $- [O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{in}(t) \right)$ term:

$$\begin{aligned} |g\rangle\langle e| |e\rangle\langle g| - |e\rangle\langle g| |g\rangle\langle e| \\ = |g\rangle\langle g| - |e\rangle\langle e| = \sigma_z \end{aligned}$$

decay: $c_1 = \sigma_- \rightarrow c^\dagger = \sigma_+ \rightarrow [O_S, c^\dagger] = [\sigma_-, \sigma_+] = e^{iHt} [|g\rangle\langle e|, |e\rangle\langle g|] e^{-iHt} = \sigma_z (t)$

$$\Rightarrow \sigma_z (t) \left(\frac{\gamma}{2} \sigma_- (t) + \sqrt{\gamma_1} b_{in,1}(t) \right)$$

$$|g\rangle\langle e| |e\rangle\langle g| - |e\rangle\langle g| |g\rangle\langle e| = |g\rangle\langle e| |e\rangle\langle e| + |e\rangle\langle e| |g\rangle\langle e| = -|g\rangle\langle e| - |g\rangle\langle e|$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\left(c_2 = \sigma_+ \rightarrow c^\dagger = \sigma_- \rightarrow [O_S, c^\dagger] = 0 \right)$$

dephasing: $c_3 = \sigma_z \rightarrow c^\dagger = \sigma_z \rightarrow [O_S, c^\dagger] = e^{iHt} [|g\rangle\langle g|, |g\rangle\langle g| - |e\rangle\langle e|] e^{-iHt} = -2 \sigma_- (t)$

$$\Rightarrow -2 \sigma_- (t) \left(\frac{\gamma}{2} \sigma_z (t) + \sqrt{\gamma_2} b_{in,2}(t) \right)$$

and the

$$\left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{in}^\dagger(t) \right) [c, O_S] \text{ term}$$

$$c_1 = \sigma_- \rightarrow [c_1, \cdot] = 0$$

$$\left(c_2 = \sigma_+ \rightarrow [\sigma_+, \sigma_-] = -[\sigma_-, \sigma_+] = -\sigma_z (t) \right)$$

$$\rightarrow - \left(\frac{\gamma}{2} \sigma_- (t) + \sqrt{\gamma_1} b_{in,1}^\dagger(t) \right) \sigma_z (t)$$

don't need this one because $T=0$

$$c_3 = \sigma_z \rightarrow [\sigma_z, \sigma_-] = -[\sigma_-, \sigma_z] = 2 \sigma_- (t) \rightarrow 2 \left(\frac{\gamma}{2} \sigma_- (t) + \sqrt{\gamma_2} b_{in,2}^\dagger(t) \right) \sigma_- (t)$$

⇒ all together

$$\sigma_2 \sigma_- = \sigma_-$$

$$\sigma_- \sigma_2 = -\sigma_-$$

$$\begin{aligned} \dot{\sigma}_-(t) &= i\Delta \sigma_-(t) - \frac{i\Omega}{2} \sigma_2(t) - \sigma_2(t) \left(\frac{\gamma_1}{2} \sigma_-(t) + \sqrt{\gamma_1} b_{in_1}(t) \right) + 2\sigma_-(t) \left(\frac{\gamma_2}{2} \sigma_2(t) + \sqrt{\gamma_2} b_{in_2}(t) \right) \\ &\quad + \left(\frac{\gamma_1}{2} \sigma_-(t) + \sqrt{\gamma_1} b_{in_1}^\dagger(t) \right) \sigma_2(t) - 2 \left(\frac{\gamma_2}{2} \sigma_2(t) + \sqrt{\gamma_2} b_{in_2}^\dagger(t) \right) \sigma_-(t) \\ &= i\Delta \sigma_-(t) - \frac{i\Omega}{2} \sigma_2(t) - \frac{\gamma_1}{2} \sigma_-(t) - \sqrt{\gamma_1} \sigma_2(t) b_{in_1}(t) - \gamma_1 \sigma_-(t) + 2\gamma_2 \sigma_-(t) b_{in_2}(t) \\ &\quad - \gamma_2 \sigma_-(t) - 2\sqrt{\gamma_2} b_{in_2}^\dagger(t) \sigma_-(t) \end{aligned}$$

$$\gamma_1 = \Gamma$$

$$\gamma_2 = \Gamma_d$$

$$\begin{aligned} \Rightarrow \dot{\sigma}_-(t) &= i\Delta \sigma_-(t) - \frac{i\Omega}{2} \sigma_2(t) - \frac{\Gamma}{2} \sigma_-(t) - \sqrt{\Gamma} \sigma_2(t) b_{in_1}(t) - 2\sqrt{\Gamma_d} \sigma_-(t) b_{in_2}(t) - 2\sqrt{\Gamma_d} b_{in_2}^\dagger(t) \sigma_-(t) \\ &= \sigma_-(t) \left(i\Delta - \frac{\Gamma}{2} - 2\sqrt{\Gamma_d} b_{in_2}(t) - 2\sqrt{\Gamma_d} b_{in_2}^\dagger(t) \right) + \sigma_2(t) \left(-\frac{i\Omega}{2} - \sqrt{\Gamma} b_{in_1}(t) \right) \end{aligned}$$

$\uparrow$ $b_{in_1}(t), \sigma_2 = 0$

$$\dot{\sigma}_+(t)$$

$$[H_S, \sigma_+] = -\hbar\Delta [\text{lexel}(t), \sigma_+(t)] + \frac{\hbar\Omega}{2} [\sigma_x(t), \sigma_+(t)] = -\hbar\Delta \sigma_+(t) + \frac{\hbar\Omega}{2} \sigma_z(t)$$

$$[\text{lexel}, \sigma_+] = |\text{lexel}\rangle\langle\text{xgl}| \cdot |\text{xgl}\rangle\langle\text{xel}| = |\text{xgl}\rangle = \sigma_+$$

$$[\sigma_x, \sigma_+] = (|\text{xgl}\rangle + |\text{gxe}\rangle)\langle\text{xgl}| - |\text{xgl}\rangle(\langle\text{xgl}| + \langle\text{gxe}|) = |\text{gxe}\rangle - |\text{lexel}\rangle = \sigma_z$$

for the $-[O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{in}(t) \right)$ term:

$$\begin{aligned} [\sigma_+, \sigma_z] &= |\text{xgl}\rangle(|\text{gxe}\rangle - |\text{lexel}\rangle) - (|\text{gxe}\rangle - |\text{lexel}\rangle)|\text{xgl}\rangle \\ &= |\text{xgl}\rangle + |\text{xgl}\rangle = 2|\text{xgl}\rangle = 2\sigma_+ \end{aligned}$$

$$c_1 = \sigma_- \rightarrow \sigma_+ = c^\dagger \rightarrow [O_S, c^\dagger] = 0$$

$$c_2 = \sigma_z$$

$$\sigma_+ \sigma_z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \sigma_+$$

$$\begin{aligned} [O_S, c^\dagger] (\dots) &= [\sigma_+, \sigma_z] \left(\frac{\Gamma_d}{2} \sigma_z + \sqrt{\Gamma_d} b_{in_2}(t) \right) = 2\sigma_+ \left(\frac{\Gamma_d}{2} \sigma_z + \sqrt{\Gamma_d} b_{in_2}(t) \right) \\ &= \Gamma_d \sigma_+(t) + 2\sqrt{\Gamma_d} \sigma_+(t) b_{in_2}(t) \end{aligned}$$

and the

$$\left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{in}^\dagger(t) \right) [c, O_S]$$

term:

$$\begin{aligned} \sigma_+ \sigma_z &= |\text{xgl}\rangle(|\text{gxe}\rangle - |\text{lexel}\rangle) = |\text{xgl}\rangle = \sigma_+ \\ \sigma_z \sigma_+ &= (|\text{gxe}\rangle - |\text{lexel}\rangle)|\text{xgl}\rangle = -|\text{xgl}\rangle \end{aligned}$$

$$c_1 = \sigma_- \rightarrow [c, O_S] = [\sigma_-, \sigma_+] = |\text{gxe}\rangle\langle\text{xgl}| - |\text{xgl}\rangle\langle\text{gxe}| = |\text{gxe}\rangle - |\text{lexel}\rangle = \sigma_z$$

$$\rightarrow \left(\frac{\Gamma}{2} \sigma_+ + \sqrt{\Gamma} b_{in_1}^\dagger(t) \right) \sigma_z(t) = \frac{\Gamma}{2} \sigma_+(t) + \sqrt{\Gamma} b_{in_1}^\dagger(t) \sigma_z(t)$$

$$c_2 = \sigma_z \rightarrow [c, O_S] = [\sigma_z, \sigma_+] = -2\sigma_+$$

$$\rightarrow -2 \left(\frac{\Gamma_\phi}{2} \sigma_z + \sqrt{\Gamma_\phi} b_{in_2}^\dagger(t) \right) \sigma_+ = \underline{\sigma_+(t) \left(\Gamma_\phi - 2\sqrt{\Gamma_\phi} b_{in_2}^\dagger(t) \right)}$$

$$\rightarrow \frac{\Gamma}{2} \sigma_+(t) + \sqrt{\Gamma} b_{in_2}^\dagger(t) \sigma_+(t) + \sigma_+(t) \left(\Gamma_\phi - 2\sqrt{\Gamma_\phi} b_{in_2}^\dagger(t) \right)$$

$$\Rightarrow \dot{\sigma}_+(t) =$$

$$-i\hbar \Delta \sigma_+(t) + \frac{\hbar\Omega}{2} \sigma_z(t) - \Gamma_\phi \sigma_+(t) - 2\sqrt{\Gamma_\phi} \sigma_+(t) b_{in_2}(t) - \frac{\Gamma}{2} \sigma_+(t) - \sqrt{\Gamma} b_{in_2}^\dagger(t) \sigma_+(t) - \sigma_+(t) \left(\Gamma_\phi - 2\sqrt{\Gamma_\phi} b_{in_2}^\dagger(t) \right)$$

$$= \sigma_+(t) \left(-i\Delta - 2\Gamma_\phi - 2\sqrt{\Gamma_\phi} b_{in_2}(t) - \frac{\Gamma}{2} + 2\sqrt{\Gamma_\phi} b_{in_2}^\dagger(t) \right) + \sigma_z(t) \left(\frac{i\Omega}{2} - \sqrt{\Gamma} b_{in_2}^\dagger(t) \right)$$

$$\sigma_2(t)$$

$$[H_S, \sigma_z] = -\hbar\Delta [\text{exel}, \sigma_z] + \frac{\hbar\Omega}{2} [\sigma_x, \sigma_z] \stackrel{=0}{=} -i\Omega\sigma_y \hbar$$

$$\frac{i}{\hbar} [H_S, \sigma_z] = \Omega\sigma_y$$

$$\sigma_+ \sigma_- = |\text{exel}\rangle \langle \text{exel}| = |\text{exel}|$$

for the $-[O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{in}(t) \right)$ term:

$$c_1 = \sigma_- \rightarrow [O_S, c^\dagger] = [\sigma_z, \sigma_+] = -2\sigma_+$$

$$\rightarrow -2\sigma_+ \left(\frac{\Gamma}{2} \sigma_- + \sqrt{\Gamma} b_{in_1}^\dagger(t) \right) = \underline{-\Gamma |\text{exel}\rangle \langle \text{exel}| - 2\sqrt{\Gamma} b_{in_1}^\dagger(t) \sigma_+(t)}$$

$$c_2 = \sigma_z \rightarrow [\sigma_z, \sigma_z] = 0 \rightarrow 0$$

and the

$$\left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{in}^\dagger(t) \right) [c, O_S] \text{ term:}$$

$$c_1 = \sigma_- \rightarrow [\sigma_-, \sigma_z] = -2\sigma_-$$

$$c_2 = \sigma_z \rightarrow 0$$

$$\rightarrow -2 \left(\frac{\Gamma}{2} \sigma_- + \sqrt{\Gamma} b_{in_1}^\dagger(t) \right) \sigma_-$$

$$= \underline{-\Gamma |\text{exel}\rangle \langle \text{exel}| - 2\sqrt{\Gamma} b_{in_1}^\dagger(t) \sigma_-(t)}$$

$$\Rightarrow \dot{\sigma}_z(t) = \Omega\sigma_y(t) + 2\Gamma |\text{exel}\rangle \langle \text{exel}| + 2\sqrt{\Gamma} (b_{in_1}^\dagger(t) \sigma_- + b_{in_1}(t) \sigma_+(t))$$

b) OBE

$$\dot{\vec{M}} = \begin{pmatrix} -\Gamma_1 & \Delta & 0 \\ -\Delta & -\Gamma_1 & -\Omega \\ 0 & \Omega & -\Gamma \end{pmatrix} \vec{M} - \begin{pmatrix} 0 \\ 0 \\ \Gamma \end{pmatrix} \quad \partial_t \vec{S} = M \vec{S} - \vec{R}$$

$$\hookrightarrow \dot{\langle \sigma_z \rangle} = \Omega \langle \sigma_y \rangle - \Gamma \langle \sigma_z \rangle - \Gamma \langle \sigma_z \rangle$$

$$\rightarrow \langle \dot{\sigma}_z(t) \rangle = \langle \Omega \sigma_y(t) + 2\Gamma |exel(t)| + 2\sqrt{\Gamma} (b_{in}^+(t) \sigma_- + b_{in}(t) \sigma_+) \rangle$$

$$= \Omega \langle \sigma_y(t) \rangle + 2\Gamma \langle |exel| \rangle + 0 \quad \langle b \rangle_{vac} = 0$$

$$= \Omega \langle \sigma_y(t) \rangle - 2\Gamma \langle \sigma_z \rangle + 2\Gamma \langle |exel| \rangle \rightarrow \text{gives the same time evolution}$$

as the Bloch equations except for this term which is 0

$$\begin{aligned} \langle \sigma_x \rangle &= \langle \sigma_+ + \sigma_- \rangle = \langle \sigma_+ \rangle + \langle \sigma_- \rangle \\ \sigma_+ &= \frac{(\sigma_x - i\sigma_y)}{2} \stackrel{2}{=} \frac{(exel + igxel - i(igxel + exel))}{2} = \frac{(exel + igxel - igxel - exel)}{2} = \sigma_+ \\ \sigma_- &= \frac{(\sigma_x + i\sigma_y)}{2} \\ &= \langle \sigma_-(t) (i\Delta - \frac{\Gamma}{2} - 2\Gamma_f + 2\sqrt{\Gamma} b_{in}(t) - 2\sqrt{\Gamma} b_{in}^+(t)) + \sigma_+(t) (-i\frac{\Omega}{2} - \sqrt{\Gamma} b_{in}(t)) \rangle \\ &+ \langle \sigma_+(t) (-i\Delta - 2\Gamma_f - 2\sqrt{\Gamma} b_{in}(t) - \frac{\Gamma}{2} + 2\sqrt{\Gamma} b_{in}^+(t)) + \sigma_-(t) (i\frac{\Omega}{2} - \sqrt{\Gamma} b_{in}(t)) \rangle \\ &= \langle \sigma_-(t) \rangle (i\Delta - \frac{\Gamma}{2} - 2\Gamma_f) - i\frac{\Omega}{2} \langle \sigma_z(t) \rangle + \langle \sigma_+(t) \rangle (-i\Delta - 2\Gamma_f - \frac{\Gamma}{2}) + \langle \sigma_z(t) \rangle i\frac{\Omega}{2} \\ &= \langle \frac{\sigma_x}{2} \rangle + i \langle \frac{\sigma_y}{2} \rangle (i\Delta - \frac{\Gamma}{2} - 2\Gamma_f) + \frac{1}{2} (\langle \sigma_x \rangle - i \langle \sigma_y \rangle) (-i\Delta - 2\Gamma_f - \frac{\Gamma}{2}) \\ &= \langle \sigma_x \rangle \left(\frac{i\Delta}{2} - \frac{i\Delta}{2} - \frac{\Gamma}{4} - \frac{\Gamma}{4} - \Gamma_f - \Gamma_f \right) + i \langle \sigma_y \rangle \left(\frac{i\Delta}{2} + \frac{i\Delta}{2} - \frac{\Gamma}{2} + \frac{\Gamma}{2} - 2\Gamma_f + 2\Gamma_f \right) \\ &= \langle \sigma_x \rangle (-\frac{\Gamma}{2} - 2\Gamma_f) + \langle \sigma_y \rangle i^2 \Delta = -\langle \sigma_x \rangle \Gamma_1 - \Delta \langle \sigma_y \rangle \end{aligned}$$

which gives the same value as the Bloch equations for $\Gamma_1 = \frac{\Gamma}{2} + 2\Gamma_f$ and $\Delta \rightarrow -\Delta$

$$\langle \sigma_y \rangle = \langle i(\sigma_+ - \sigma_-) \rangle = i \left(\langle \sigma_-(t) \rangle (i\Delta - \frac{\Gamma}{2} - 2\Gamma_f) - i\frac{\Omega}{2} \langle \sigma_z(t) \rangle - \langle \sigma_+(t) \rangle (-i\Delta - 2\Gamma_f - \frac{\Gamma}{2}) - \langle \sigma_z(t) \rangle i\frac{\Omega}{2} \right)$$

$$\begin{aligned} &= i \left(\langle \sigma_x + i\sigma_y \rangle (i\Delta - \frac{\Gamma}{2} - 2\Gamma_f) - 2i\Omega \langle \sigma_z(t) \rangle - \langle \sigma_x - i\sigma_y \rangle (-i\Delta - 2\Gamma_f - \frac{\Gamma}{2}) \right) \\ &= \langle \sigma_y \rangle \left(i\frac{1}{2} \left[(i\Delta - \frac{\Gamma}{2} - 2\Gamma_f) - (-i\Delta - 2\Gamma_f - \frac{\Gamma}{2}) \right] \right) + i \langle \sigma_y \rangle \left(i\frac{1}{2} \left[(i\Delta - \frac{\Gamma}{2} - 2\Gamma_f) + (-i\Delta - 2\Gamma_f - \frac{\Gamma}{2}) \right] \right) + \Omega \langle \sigma_z \rangle \\ &= \langle \sigma_x \rangle \left(i\frac{1}{2} \cdot 2i\Delta \right) + i \cdot \langle \sigma_y \rangle \left(\frac{1}{2} (-4\Gamma_f - \Gamma) \right) + \Omega \langle \sigma_z \rangle \\ &= -\langle \sigma_x \rangle \Delta + \langle \sigma_y \rangle \Gamma_1 + \Omega \langle \sigma_z \rangle \end{aligned}$$

which gives the same evolution as the Bloch equations (up to signs and

$$c) d_t(\sigma_+ \sigma_-) = \dot{\sigma}_+ \sigma_- + \sigma_+ \dot{\sigma}_-$$

$$\sigma_z \sigma_- = |ex\rangle \langle ex| = |ex\rangle \langle ex|$$

$$\sigma_+ \sigma_z = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\sigma_z \sigma_- = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \sigma_+$$

$$\dot{\sigma}_+ \sigma_- = \sigma_+(t) \left(-i\Delta - 2T\phi - 2\sqrt{T\phi} b_{in_2}(t) - \frac{T}{2} + 2\sqrt{T\phi} b_{in_2}^\dagger(t) \right) + \sigma_z(t) \left(i\frac{\Omega}{2} - \sqrt{T} b_{in_1}^\dagger(t) \right) \sigma_-(t)$$

$$\sigma_+(t) \sigma_-(t) \left(-i\Delta - 2T\phi - 2\sqrt{T\phi} b_{in_2}(t) - \frac{T}{2} + 2\sqrt{T\phi} b_{in_2}^\dagger(t) \right) + \sigma_z(t) \sigma_-(t) \left(i\frac{\Omega}{2} - \sqrt{T} b_{in_1}^\dagger(t) \right)$$

$$= |ex\rangle \langle ex| \left(-i\Delta - 2T\phi - 2\sqrt{T\phi} b_{in_2}(t) - \frac{T}{2} + 2\sqrt{T\phi} b_{in_2}^\dagger(t) \right) + \sigma_-(t) \left(i\frac{\Omega}{2} - \sqrt{T} b_{in_1}^\dagger(t) \right)$$

$$\sigma_+ \dot{\sigma}_- = \sigma_+ \left(\sigma_-(t) \left(i\Delta - \frac{T}{2} - 2T\phi + 2\sqrt{T\phi} b_{in_2}(t) - 2\sqrt{T\phi} b_{in_2}^\dagger(t) \right) + \sigma_z(t) \left(-i\frac{\Omega}{2} - \sqrt{T} b_{in_1}^\dagger(t) \right) \right)$$

$$= |ex\rangle \langle ex| \left(i\Delta - \frac{T}{2} - 2T\phi + 2\sqrt{T\phi} b_{in_2}(t) - 2\sqrt{T\phi} b_{in_2}^\dagger(t) \right) + \sigma_+(t) \left(-i\frac{\Omega}{2} - \sqrt{T} b_{in_1}^\dagger(t) \right)$$

$$\Rightarrow I + II = |ex\rangle \langle ex| \left(-4T\phi - T \right) + \sigma_-(t) \left(i\frac{\Omega}{2} - \sqrt{T} b_{in_1}^\dagger(t) \right) + \sigma_+(t) \left(-i\frac{\Omega}{2} - \sqrt{T} b_{in_1}^\dagger(t) \right)$$

$$= |ex\rangle \langle ex| \left(-4T\phi - T \right) + i\frac{\Omega}{2} \left(\sigma_-(t) - \sigma_+(t) \right) \left(-\sigma_- \sqrt{T} b_{in_1}^\dagger(t) - \sigma_+ \sqrt{T} b_{in_1}^\dagger(t) \right)$$

$$= |ex\rangle \langle ex| \left(-4T\phi - T \right) - \frac{\Omega}{2} \sigma_y - \sigma_- \sqrt{T} b_{in_1}^\dagger(t) - \sigma_+ \sqrt{T} b_{in_1}^\dagger(t)$$

$$\partial_t |ex\rangle \langle ex| [H_S, |ex\rangle \langle ex|] = -\hbar\Delta [|ex\rangle \langle ex|, |ex\rangle \langle ex|] + \frac{\hbar\Omega}{2} [\sigma_x, |ex\rangle \langle ex|]$$

$$= 0 + \frac{\hbar\Omega}{2} \left((|ex\rangle \langle ex| + |gx\rangle \langle ex|) |ex\rangle \langle ex| - |ex\rangle \langle ex| (|ex\rangle \langle ex| + |gx\rangle \langle ex|) \right)$$

$$= \frac{\hbar\Omega}{2} (|gx\rangle \langle ex| - |ex\rangle \langle gx|) = \frac{\hbar\Omega}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = -\frac{\hbar\Omega}{2} \hat{\sigma}_y$$

$$\Rightarrow \frac{i}{\hbar} [H_S, |ex\rangle \langle ex|] = -\frac{\Omega}{2} \hat{\sigma}_y$$

$$- [O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{in}(t) \right) \quad \text{term:}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$c_- = \sigma_- : [O_S, c^\dagger] = [|ex\rangle \langle ex|, \sigma^+] = \sigma^+$$

$$\Rightarrow \sigma^+ \left(\frac{T}{2} \sigma_- + \sqrt{T} b_{in_1}(t) \right) = \frac{T}{2} |ex\rangle \langle ex| + \sigma_+ \sqrt{T} b_{in_1}(t)$$

$$c_z = \sigma_z \rightarrow [., -] = 0$$

$$\left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{in}^\dagger(t) \right) [c, O_S] \quad \text{term}$$

$$[c, O_S] = [\sigma_-, |ex\rangle \langle ex|] = |gx\rangle \langle ex| |ex\rangle \langle ex| = |gx\rangle \langle ex| = \sigma_-$$

$$\rightarrow \left(\frac{T}{2} \sigma_+ + \sqrt{T} b_{in_1}^\dagger(t) \right) \sigma_- = \frac{T}{2} |ex\rangle \langle ex| + \sqrt{T} b_{in_1}^\dagger(t) \sigma_-$$

⇒ together

$$\partial_t |eX\rangle = -\frac{\Omega}{2} \sigma_y - \Gamma |eX\rangle - \sigma_+ \sqrt{\Gamma} b_{in}(t) - \sqrt{\Gamma} b_{in}^\dagger(t) \sigma_-$$

which is the same expression as above, except for the T_ϕ -term. I assume I had a wrong sign here which should have made that one cancel

22. Cascaded quantum systems (2P)

Consider two harmonic oscillators, which are coupled to a unidirectional waveguide at positions z_1 and z_2 , as shown in Fig. 1. The Hamiltonian for the full system is given by

$$H = \sum_{i=1}^2 \hbar \omega_0 a_i^\dagger a_i + \int_{\omega_0 - \Delta}^{\omega_0 + \Delta} d\omega \hbar \omega b_\omega^\dagger b_\omega + H_{\text{int}}, \quad (5)$$

where $[b_\omega, b_{\omega'}^\dagger] = \delta(\omega - \omega')$ and the interaction Hamiltonian is

$$H_{\text{int}} = i \sum_{i=1}^2 \sqrt{\frac{\gamma_i}{2\pi}} \left[b^\dagger(z_i) a_i - b(z_i) a_i^\dagger \right], \quad b(z, t) = \frac{1}{\sqrt{2\pi}} \int d\omega e^{i\omega z/c} b_\omega(t). \quad (6)$$

(a) Derive the QLEs for the operators a_1 and a_2 and express them in terms of the incoming field

$$b_{\text{in}}(t) = \lim_{\epsilon \rightarrow 0} b(z_1 - \epsilon, t). \quad (7)$$

(b) Express the outgoing field $b(z, t)$ at position $z > z_2$ in terms of the $a_i(t)$ and $b_{\text{in}}(t)$.

a)

Quantum Langevin Equation (QLE):

$$\dot{O}_S = \frac{i}{\hbar} [H_S, O_S] - [O_S, c^\dagger] \left(\frac{\gamma}{2} c + \sqrt{\gamma} b_{\text{in}}(t) \right) - \left(\frac{\gamma}{2} c^\dagger + \sqrt{\gamma} b_{\text{in}}^\dagger(t) \right) [c, O_S]$$

$$\{a^\dagger a, a\} = a^\dagger \{a, a\} + \{a^\dagger, a\} a$$

$$H_S = \sum_{i=1}^2 \hbar \omega_0 \hat{a}_i^\dagger \hat{a}_i = \hbar \omega_0 (\hat{a}_1^\dagger \hat{a}_1 + \hat{a}_2^\dagger \hat{a}_2)$$

$$\frac{i}{\hbar} [H_S, \hat{a}_i] = i \omega_0 \{a_i^\dagger a_i, a_i\} = i \omega_0 \hat{a}_i$$

$$b_{\text{in}}(t) = \lim_{\epsilon \rightarrow 0} b(z_1 - \epsilon, t)$$

$$b_{\text{out}}(t) = b_{\text{in}}(t) + \sqrt{\frac{\gamma}{2}} a_1(t)$$

a_1

$$c_i = a_i \rightarrow [c, O_S] = 0$$

$$[O_S, c^\dagger] = [a_i, a_i^\dagger] = -1$$

$$\rightarrow + \left(\frac{\gamma}{2} a_1(t) + \sqrt{\gamma} b_{\text{in}}(t) \right)$$

why is $[b_{\text{in}}, O_S] \neq 0$?

$$e^{iHt} b_{\text{in}}(0) e^{-iHt} e^{iHt} O_S e^{-iHt} - \dots$$

$$= e^{iHt} \underbrace{\{b_{\text{in}}(0), O_S(0)\}}_{=0} e^{-iHt} = 0 ??$$

$$a_2: c_2 = a_2 \rightarrow [c, c_S] = 0$$

$$\Rightarrow [c_S, c^\dagger] = 1$$

$$\rightarrow \left(\frac{\sigma}{2} a_2(t) + \sqrt{\gamma} b_{in_2}(t) \right)$$

$$\Rightarrow \dot{a}_1(t) = (i\omega_0 - \frac{\sigma}{2}) a_1(t) - \sqrt{\gamma} b_{in}(t)$$

$$\dot{a}_2(t) = (i\omega_0 - \frac{\sigma}{2}) a_2(t) - \sqrt{\gamma} (b_{in}(t) + \sqrt{\frac{\sigma}{2}} a_1(t))$$

b)

(b) Express the outgoing field $b(z, t)$ at position $z > z_2$ in terms of the $a_1(t)$ and $b_{in}(t)$.

$b(z, t)$ for $z > z_2$

$$= b_{out}^2(t) = b_{in}^2(t) + \sqrt{\frac{\sigma}{2}} a_2(t) = b_{in}(t) + \sqrt{\frac{\sigma}{2}} (a_1(t) + a_2(t))$$